Primitive Root

① Show that 2 is a primitive root modulo 11

⇒ Given,

A number g is a primitive root modulo 11

Since 11 is prime, we need the order of 2 modulo 11 to be $\phi(11) = 10$.

Compute powers of 2 modulo 11

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16 \equiv 5 \mod (11)$

$2^5 = 2^4 \cdot 2 = 5 \cdot 2 = 10$

$2^6 = 10 \cdot 2 = 20 \equiv 9 \mod (11)$

$2^7 = 9 \cdot 2 = 18 \equiv 7 \mod (11)$

$2^8 = 7 \cdot 2 = 14 \equiv 3 \mod (11)$

$2^9 = 3 \cdot 2 = 6$

$2^{10} = 6 \cdot 2 = 12 \equiv 1 \mod (11)$

So, we reached 1 first at exponent 10. So, the order of 2 mod 11 is $10 = \phi(11)$

Therefore 2 is a primitive root modulo 11

② How many incongruent primitive root does 14 have?

⇒ Primitive roots exists for $n = 2, 4, p^k$ or $2p^k$ with odd prime p. Since $14 = 2 \cdot 7$, primitive root exists.

The number of incongruent primitive roots modulo. compute,

$$\phi(14) = \phi(2)\,\phi(7) = 1 \cdot 6 = 6$$

So number of primitive roots $= \phi(6) = 2$ (Ans)

③ ⓐ Show that ~~orders~~

$$\mathrm{Ord}_n(a) = \mathrm{Ord}_n(a^{-1})$$

Let. $\mathrm{Ord}_n(a) = k$

$\therefore a^k \equiv 1 \mod (n)$

$\therefore (a^k)^{-1} \equiv 1^{-1} \mod (n)$

$\therefore (a^{-1})^k \equiv 1 \mod (n)$

That means the order of a^{-1} divides k

Hence the two orders divide each other. So the are equal.

$\therefore \mathrm{Ord}_n(a) = \mathrm{Ord}_n(a^{-1})$

(b) Yes.

If a is a primitive root mod. n,

then $\text{Ord}_n(a) = \phi(n)$

From part (a),

$$\text{ord}_n(a^{-1}) = \text{ord}_n(a) = \phi(n)$$

So, a^{-1} also has order $\phi(n)$

Therefore, a^{-1} is also a primitive root modulo n.